

HO CHI MINH CITY NATIONAL UNIVERSITY
UNIVERSITY OF NATURAL SCIENCES
FACULTY OF INFORMATION TECHNOLOGY



DATA VISUALIZATION USING TABLEAU

Instructor:

Nguyễn Ngọc Minh Châu

GROUP INFORMATION

<i>Group code</i>	<i>MSSV</i>	<i>Full name</i>	<i>Note</i>
22HTTT Group 11	22127107	Nguyễn Thế Hiển	Team Leader
	22127260	Bùi Công Mậu	
	22127355	Nguyễn Trần Đại Quốc	
	22127400	Thái Hữu Thọ	

NOTE: THE EXAMPLES ARE ONLY USED TO DESCRIPTION THE CORRESPONDING SECTION, NOT TO GUIDE STUDENTS TO DO THE EXERCISE.

1. Dataprofiling

1.1 Data preprocessing

Data provided is from kaggle: <https://www.kaggle.com/datasets/shivamb/netflix-shows>.

```
show_id,type,title,director,cast,country,date_added,release_year,rating,duration,listed_in,description
s1,Movie,Dick Johnson Is Dead,Kirsten Johnson,United States,"September 25, 2021",2020,PG-13,90 min,Documentaries,"As her father n
s2,TV Show,Blood & Water,,,"Ama Qamata, Khosi Ngema, Gail Mababane, Thabang Molaba, Dillon Windvogel, Natasha Thahane, Arno Greeff,
s3,TV Show,Ganglands,Julien Leclercq,"Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabiha Akkari, Sofia Lesaffre, Salim Kechiouche, N
s4,TV Show,Jailbirds New Orleans,,,,,"September 24, 2021",2021,TV-MA,1 Season,"Docuseries, Reality TV","Feuds, flirtations and toil
s5,TV Show,Kota Factory,,,"Mayur More, Jitendra Kumar, Ranjan Raj, Alam Khan, Ahsaas Channa, Revathi Pillai, Urvi Singh, Arun Kumar
s6,TV Show,Midnight Mass,Mike Flanagan,"Kate Siegel, Zach Gilford, Hamish Linklater, Henry Thomas, Kristin Lehman, Samantha Sloan
s7,Movie,My Little Pony: A New Generation,"Robert Cullen, José Luis Ucha",Vanessa Hudgens, Kimiko Glenn, James Marsden, Sofia Car
s8,Movie,Sankofa,Haile Gerima,"Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra Duah, Nick Medley, Mutabaruka, Afemo Omilami, Reggie C
s9,TV Show,The Great British Baking Show,Andy Devonshire,"Mel Giedroyc, Sue Perkins, Mary Berry, Paul Hollywood",United Kingdom,"S
s10,Movie,The Starling,Theodore Melfi,"Melissa McCarthy, Chris O'Dowd, Kevin Kline, Timothy Olyphant, Daveed Diggs, Skyler Gisondo
s11,TV Show,"Vendetta: Truth, Lies and The Mafia",,,,,,"September 24, 2021",2021,TV-MA,1 Season,"Crime TV Shows, Docuseries, Interna
s12,TV Show,Bangkok Breaking,Kongkiat Komesiri,"Sukollawat Kanarot, Sushar Manaying, Pavarit Mongkolpisit, Sahajak Boonthanakit, S
s13,Movie,Je Suis Karl,Christian Schwochow,"Luna Wedler, Jannis Niewöhner, Milan Peschel, Edin Hasanović, Anna Fialová, Marlon Boe
s14,Movie,Confessions of an Invisible Girl,Bruno Garotti,"Klara Castanho, Lucca Picon, Júlia Gomes, Marcus Bessa, Kiria Malheiros,
s15,TV Show,Crime Stories: India Detectives,,,,,"September 22, 2021",2021,TV-MA,1 Season,"British TV Shows, Crime TV Shows, Docuser
s16,TV Show,Dear White People,,,"Logan Browning, Brandon P. Bell, DeRon Horton, Antoinette Robertson, John Patrick Amedori, Ashley
s17,Movie,Europe's Most Dangerous Man: Otto Skorzeny in Spain,"Pedro de Echave García, Pablo Azorín Williams",,"September 22, 202
s18,TV Show,Falsa identidad,,,"Luis Ernesto Franco, Camila Sodi, Sergio Goyri, Samadhi Zendejas, Eduardo Yáñez, Sonya Smith, Alejan
s19,Movie,Intrusion,Adam Salky,"Freida Pinto, Logan Marshall-Green, Robert John Burke, Megan Elisabeth Kelly, Sarah Minnich, Hayes
s20,TV Show,Jaguar,,,"Blanca Suárez, Iván Marcos, Óscar Casas, Adrián Lastra, Francesc Garrido, Stefan Weinert, Julia Möller, Alici
s21,TV Show,Monsters Inside: The 24 Faces of Billy Milligan,Olivier Megaton,,,"September 22, 2021",2021,TV-14,1 Season,"Crime TV S
s22,TV Show,Resurrection: Ertugrul,,,"Engin Altan Düzyatan, Serdar Gökhan, Hülya Darcan, Kaan Taşaner, Esra Bilgiç, Osman Soykut, S
s23,Movie,Avvai Shanmughi,K.S. Ravikumar,"Kamal Hassan, Meena, Gemini Ganesan, Heera Rajgopal, Nassar, S.P. Balasubrahmanyam",,"Se
s24,Movie,Go! Go! Cory Carson: Chrissy Takes the Wheel,"Alex Woo, Stanley Moore",,"Maisie Benson, Paul Killam, Kerry Gudjohnsen, AC
s25,Movie,Jeans,S. Shankar,"Prashanth, Aishwarya Rai Bachchan, Sri Lakshmi, Nassar",India,"September 21, 2021",1998,TV-14,166 min,
s26,TV Show,Love on the Spectrum,,Brooke Satchwell,Australia,"September 21, 2021",2021,TV-14,2 Seasons,"Docuseries, International
s27,Movie,Minsara Kanavu,Rajiv Menon,"Arvind Swamy, Kajol, Prabhu Deva, Nassar, S.P. Balasubrahmanyam, Girish Karnad",,"September
s28,Movie,Grown Ups,Dennis Dugan,"Adam Sandler, Kevin James, Chris Rock, David Spade, Rob Schneider, Salma Hayek, Maria Bello, May
s29,Movie,Dark Skies,Scott Stewart,"Keri Russell, Josh Hamilton, J.K. Simmons, Dakota Goyo, Kadan Rockett, L.J. Benet, Rich Hutchm
s30,Movie,Paranoia,Robert Luketic,"Liam Hemsworth, Gary Oldman, Amber Heard, Harrison Ford, Lucas Till, Embeth Davidtz, Julian McM
s31,Movie,Ankahi Kahaniya,"Ashwiny Iyer Tiwari, Abhishek Chaubey, Saket Chaudhary",,"Abhishek Banerjee, Rinku Rajguru, Delzad Hiwal
s32,TV Show,Chicago Party Aunt,,Lauren Ash, Rory O'Malley, RuPaul Charles, Jill Talley, Ike Barinholtz, Jon Barinholtz, Matthew C
s33,TV Show,Sex Education,,Asa Butterfield, Gillian Anderson, Ncuti Gatwa, Emma Mackey, Connor Swindells, Kedar Williams-Stirling
s34,TV Show,Squid Game,,Lee Jung-jae, Park Hae-soo, Wi Ha-jun, Oh Young-soo, Jung Ho-yeon, Heo Sung-tae, Kim Joo-ryoung, Tripathi
s35,TV Show,Tayo and Little Wizards,,Dami Lee, Jason Lee, Bonnie Catherine Han, Jennifer Waeschler, Nancy Kim,,,"September 17, 202
s36,Movie,The Father Who Moves Mountains,Daniel Sandu,"Adrian Titieni, Elena Parea, Judith State, Valeriu Andriuşă, Tudor Smoleanu
s37,Movie,The Stronghold,Cédric Jimenez,"Gilles Lellouche, Karim Leklou, François Civil, Adèle Exarchopoulos, Kenza Fortas, Cyril
s38,TV Show,Angry Birds,,Antti Pääkkönen, Heljä Heikkinen, Lynne Guaglione, Pasi Ruohonen, Rauno Ahonen",Finland,"September 16, 2
```

File netflix_titles.csv

The data consists of 8807 tuples (without header), with a total of 11 attributes described as follows:

Properties	Describe
Show_id	A code that identifies a movie or series. Each movie or series has a unique code.
Type	The type of the movie or series. For example, it includes values like "Movie" or "TV Show".
Title	Movie name.
Director	Director of a film or series.
Cast	List of actors who played the main character in the film or series.

Country	List of countries where the movie or series was filmed.
Date_added	The date the movie or series was released on Netflix
Release_year	The year the movie or series was released.
Rating	The rating of a movie or series based on audience reviews. Often given in conjunction with an age rating (e.g. PG-13, R).
Duration	Movie duration in minutes. Total time of a movie or a whole series.
Listed_in	Genre of movie, series.
Description	A brief description of the movie or series. A quick summary of the main plot of the movie.

The data is considered raw data, there are quite a lot of data with missing values.:

```

#      Column      Non-Null Count  Dtype
---  -
0     show_id      8807 non-null    object
1     type         8807 non-null    object
2     title        8807 non-null    object
3     director     6173 non-null    object
4     cast         7982 non-null    object
5     country      7976 non-null    object
6     date_added   8797 non-null    object
7     release_year 8807 non-null    int64
8     rating       8803 non-null    object
9     duration     8804 non-null    object
10    listed_in     8807 non-null    object
11    description   8807 non-null    object
dtypes: int64(1), object(11)

```

Based on the above information, we perform the data preprocessing process:

+ For data types in the form of numbers (float), fill in the blank cells with the mean

value.

+ For data types in the form of objects, fill in the blank cells with the value "Unknown" or "No information" for identification. In these cases, it is not possible to fill in specific values because it is easy to distort the information during data processing.

After the preprocessing process, we get clean data in the file

"netflix_titles_cleaned.csv":

```
show_id,type,title,director,cast,country,date_added,release_year,rating,duration,listed_in,description
s1,Movie,Dick Johnson Is Dead,Kirsten Johnson,Không có thông tin,United States,"September 25, 2021",2020,PG-13,90 min,Documentaries,"As her father nears the end of his life, fil
s2,TV Show,Blood & Water,Không có thông tin,"Ama Qamata, Khosi Ngema, Gail Mablane, Thabang Molaba, Dillon Windvogel, Natasha Thahane, Arno Greeff, Xolile Tshabalala, Getmore S
s3,TV Show,Ganglands,Julien Leclercq,"Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabiha Akkari, Sofia Lesaffre, Salim Kechiouche, Noureddine Farihi, Geert Van Rangelberg, Bakary
s4,TV Show,Jailbirds New Orleans,Không có thông tin,Không có thông tin,Không xác định,"September 24, 2021",2021,TV-MA,1 Season,"Docuseries, Reality TV","Feuds, flirtations and t
s5,TV Show,Kota Factory,Không có thông tin,"Mayur More, Jitendra Kumar, Ranjan Raj, Alam Khan, Ahsaas Channa, Revathi Pillai, Urvi Singh, Arun Kumar",India,"September 24, 2021",
s6,TV Show,Midnight Mass,Mike Flanagan,"Kate Siegel, Zach Gilford, Hamish Linklater, Henry Thomas, Kristin Lehman, Samantha Sloan, Igby Rigney, Rahul Kohli, Annarah Cymone, Anr
s7,Movie,My Little Pony: A New Generation,"Robert Cullen, José Luis Ucha",,"Vanessa Hudgens, Kimiko Glenn, James Marsden, Sofia Carson, Liza Koshy, Ken Jeong, Elizabeth Perkins,
s8,Movie,Sankofa,Haile Gerima,"Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra Duah, Nick Medley, Mutabaruka, Afemo Omilami, Reggie Carter, Mzuri",,"United States, Ghana, Burkina Fe
s9,TV Show,The Great British Baking Show,Andy Devonshire,"Mel Giedroyc, Sue Perkins, Mary Berry, Paul Hollywood",United Kingdom,"September 24, 2021",2021,TV-14,9 Seasons,"Britis
s10,Movie,The Starling,Theodore Melfi,"Melissa McCarthy, Chris O'Dowd, Kevin Kline, Timothy Olyphant, Daveed Diggs, Skyler Gisondo, Laura Harrier, Rosalind Rao, Kimberly Quinn,
s11,TV Show,"Vendetta: Truth, Lies and The Mafia",Không có thông tin,Không có thông tin,Không xác định,"September 24, 2021",2021,TV-MA,1 Season,"Crime TV Shows, Docuseries, Inte
s12,TV Show,Bangkok Breaking,Kongkiat Komesiri,"Sukollawat Kanarot, Sushar Manaying, Pavarit Mongkolpisit, Sahajak Boonthanakit, Suthipongse Thatphithakul, Bhasaworn Bawronkire
s13,Movie,Je Suis Karl,Christian Schwochow,"Luna Medler, Jannis Niewöhner, Milan Peschel, Edin Hasanović, Anna Fialová, Marlon Boess, Victor Boccard, Fleur Geffrier, Aziz Dyab,
s14,Movie,Confessions of an Invisible Girl,Bruno Garotti,"Klara Castanho, Lucca Picon, Júlia Gomes, Marcus Bessa, Kiria Malheiros, Fernanda Concon, Gabriel Lima, Caio Cabral, Le
s15,TV Show,Crime Stories: India Detectives,Không có thông tin,Không có thông tin,Không xác định,"September 22, 2021",2021,TV-MA,1 Season,"British TV Shows, Crime TV Shows, Docu
s16,TV Show,Dear White People,Không có thông tin,"Logan Browning, Brandon P. Bell, DeRon Horton, Antoinette Robertson, John Patrick Amedori, Ashley Blaine Featherson, Marque Ric
s17,Movie,Europe's Most Dangerous Man: Otto Skorzeny in Spain,"Pedro de Echave García, Pablo Azorín Williams",,"Không có thông tin,Không xác định,"September 22, 2021",2020,TV-MA,6
s18,TV Show,Falsa identidad,Không có thông tin,"Luis Ernesto Franco, Camila Sodi, Sergio Goyri, Samadhi Zendejas, Eduardo Yáñez, Sonya Smith, Alejandro Camacho, Azela Robinson,
s19,Movie,Intrusion,Adam Salky,"Freida Pinto, Logan Marshall-Green, Robert John Burke, Megan Elisabeth Kelly, Sarah Minnich, Hayes Hargrove, Mark Silvestsen, Brandon Fierro, Antc
s20,TV Show,Jaguar,Không có thông tin,"Blanca Suárez, Iván Marcos, Óscar Casas, Adrián Lastra, Francesc Garrido, Stefan Weinert, Julia Möller, Alicia Chojnowski",,"Không xác định,
s21,TV Show,Monsters Inside: The 24 Faces of Billy Milligan,Olivier Megaton,Không có thông tin,Không xác định,"September 22, 2021",2021,TV-14,1 Season,"Crime TV Shows, Docuserie
s22,TV Show,Resurrection: Ertugrul,Không có thông tin,"Engin Altan Düzyatan, Serdar Gökhan, Hülya Darcan, Kaan Tazaner, Esra Bilgiç, Osman Soykut, Serdar Deniz, Cengiz Coskun, F
s23,Movie,Avvai Shanmughi,K.S. Ravikumar,"Kamal Hassan, Meena, Gemini Ganesan, Heera Rajgopal, Nassar, S.P. Balasubrahmanyam",,"Không xác định,"September 21, 2021",1996,TV-PG,161
s24,Movie,Go! Go! Cory Carson: Chrissy Takes the Wheel,"Alex Woo, Stanley Moore",,"Maisie Benson, Paul Killam, Kerry Gudjohnsen, AC Lim",,"Không xác định,"September 21, 2021",2021,
s25,Movie,Jeans,S. Shankar,"Prashanth, Aishwarya Rai Bachchan, Sri Lakshmi, Nassar",,"India,"September 21, 2021",1998,TV-14,166 min,"Comedies, International Movies, Romantic Movie
s26,TV Show,Love on the Spectrum,Không có thông tin,Brooke Satchwell,Australia,"September 21, 2021",2021,TV-14,2 Seasons,"Docuseries, International TV Shows, Reality TV","Findir
s27,Movie,Minsara Kanavu,Rajiv Menon,"Arvind Swamy, Kajol, Prabhu Deva, Nassar, S.P. Balasubrahmanyam, Girish Karnad",,"Không xác định,"September 21, 2021",1997,TV-PG,147 min,"Com
s28,Movie,Grown Ups,Dennis Dugan,"Adam Sandler, Kevin James, Chris Rock, David Spade, Rob Schneider, Salma Hayek, Maria Bello, Maya Rudolph, Colin Quinn, Tim Meadows, Joyce Van
s29,Movie,Dark Skies,Scott Stewart,"Keri Russell, Josh Hamilton, J.K. Simmons, Dakota Goyo, Kadan Rockett, L.J. Benet, Rich Hutchman, Myndy Crist, Annie Thurman, Jake Brennan",,"l
s30,Movie,Paranoia,Robert Luketic,"Liam Hemsworth, Gary Oldman, Amber Heard, Harrison Ford, Lucas Till, Embeth Davidtz, Julian McMahon, Josh Holloway, Richard Dreyfuss, Angela S
s31,Movie,Ankahi Kahaniya,"Ashwiny Iyer Tiwari, Abhishek Chaubey, Saket Chaudhary",,"Abhishek Banerjee, Rinku Rajguru, Delzad Hiwale, Kunal Kapoor, Zoya Hussain, Nikhil Dwivedi,
s32,TV Show,Chicago Party Aunt,Không có thông tin,"Lauren Ash, Rory O'Malley, RuPaul Charles, Jill Talley, Ike Barinholtz, Jon Barinholtz, Matthew Craig, Bob Odenkirk, Mike Hage
s33,TV Show,Sex Education,Không có thông tin,"Asa Butterfield, Gillian Anderson, Ncuti Gatwa, Emma Mackey, Connor Swindells, Kedar Williams-Stirling, Alistair Petrie",,"United Kir
s34,TV Show,Squid Game,Không có thông tin,"Lee Jung-jae, Park Hae-soo, Wi Ha-jun, Oh Young-soo, Jung Ho-yeon, Heo Sung-tae, Kim Joo-ryoung, Tripathi Anupam, You Seong-joo, Lee Y
s35,TV Show,Tayo and Little Wizards,Không có thông tin,"Dami Lee, Jason Lee, Bonnie Catherine Han, Jennifer Waescher, Nancy Kim",,"Không xác định,"September 17, 2021",2020,TV-Y7,1
s36,Movie,The Father Who Moves Mountains,Daniel Sandu,"Adrian Titieni, Elena Porea, Judith State, Valeriu Andriută, Tudor Smoleanu, Virgil Aioanei, Radu Botar, Petronela Grigore
s37,Movie,The Stronghold,Cédric Jimenez,"Gilles Lellouche, Karim Leklou, François Civil, Adèle Exarchopoulos, Kenza Fortas, Cyril Lecomte, Michaël Abiteboul, Idir Azougli, Vince
s38,TV Show,Angry Birds,Không có thông tin,"Antti Pääkkönen, Heljä Heikkonen, Lynne Guaglione, Pasi Ruohonen, Rauno Aho",,"Finland,"September 16, 2021",2018,TV-Y7,1 Season,"Kic
s39,Movie,Birth of the Dragon,George Nolfi,"Billy Magnussen, Ron Yuan, Qu Jingjing, Terry Chen, Vanness Wu, Jin Xing, Philip Ng, Xia Yu, Yu Xia",,"China, Canada, United States",,"
s40,TV Show,Chhota Bheem,Không có thông tin,"Vatsal Dubey, Julie Tejwani, Rupa Bhimani, Jigna Bhardwaj, Rajesh Kava, Mousam, Swapnil",,"India,"September 16, 2021",2021,TV-Y7,3 See
s41,TV Show,He-Man and the Masters of the Universe,Không có thông tin,"Yuri Lowenthal, Kimberly Brooks, Antony Del Rio, Trevor Devall, Ben Diskin, Grey Griffin, David Kaye, Tom
s42,Movie,Jaws,Steven Spielberg,"Roy Scheider, Robert Shaw, Richard Dreyfuss, Lorraine Gary, Murray Hamilton, Carl Gottlieb, Jeffrey Kramer, Susan Backlinie, Jonathan Filley, Te
s43,Movie,Jaws 2,Jeannot Szwarc,"Roy Scheider, Lorraine Gary, Murray Hamilton, Joseph Mascolo, Jeffrey Kramer, Collin Wilcox Paxton, Ann Dusenberry, Mark Gruner, Barry Coe, Susa
s44,Movie,Jaws 3,Joe Alves,"Dennis Quaid, Bess Armstrong, Simon MacCorkindale, Louis Gossett Jr., John Putch, Lea Thompson, P.H. Moriarty, Dan Blasko, Liz Morris, Lisa Maurer",,"l
s45,Movie,Jaws: The Revenge,Joseph Sargent,"Lorraine Gary, Lance Guest, Mario Van Peebles, Karen Young, Michael Caine, Judith Barsi, Mitchell Anderson, Lynn Whitfield",,"United St
s46,Movie,My Heroes Were Cowboys,Tyler Greco,Không có thông tin,Không xác định,"September 16, 2021",2021,PG,23 min,Documentaries,"Robin Wiltshire's painful childhood was rescuec
s47,Movie,Safe House,Daniel Espinosa,"Denzel Washington, Ryan Reynolds, Vera Farmiga, Brendan Gleeson, Sam Shepard, Rubén Blades, Nora Arnezeder, Robert Patrick, Liam Cunningham
s48,TV Show,The Smart Money Woman,Bunmi Ajakaiye,"Osas Ighodaro, Ini Dima-Okojie, Kemi Lala Akindoju, Toni Tones, Ebenezer Eno, Eso Okolocha DiKe, Patrick Diabuah, Karibi Fubare
s49,Movie,Training Day,Antoine Fuqua,"Denzel Washington, Ethan Hawke, Scott Glenn, Tom Berenger, Harris Yulin, Raymond J. Barry, Cliff Curtis, Dr. Dre, Snoop Dogg, Macy Gray, Ev
```

File netflix_titles_cleaned.csv

1.2 Profiling all data

Proceed to profile all the data and get the following results::

		Field Name	NULL	Missing	Actual	Completeness (%)	Cardinality	Uniqueness (%)	Distinctness (%)
0	0	show_id	0	0	8807	100.00	8807	100.00	100.00
1	1	type	0	0	8807	100.00	2	0.02	0.02
2	2	title	0	0	8807	100.00	8806	99.99	99.99
3	3	director	2634	0	6173	70.09	4528	51.41	51.41
4	4	cast	825	0	7982	90.63	7692	87.34	87.34
5	5	country	831	0	7976	90.56	748	8.49	8.49
6	6	date_added	10	0	8797	99.89	1767	20.06	20.06
7	7	release_year	0	0	8807	100.00	74	0.84	0.84
8	8	rating	4	0	8803	99.95	17	0.19	0.19
9	9	duration	3	0	8804	99.97	220	2.50	2.50
10	10	listed_in	0	0	8807	100.00	514	5.84	5.84
11	11	description	0	0	8807	100.00	8775	99.64	99.64

Comments:

- Before preprocessing the data, the attributes show_id, type, title, release_year, listed_in, description have completeness of 100%.78
- date_added, rating, duration all have completeness > 99.8% with only a few missing values, easy to preprocess.
- duration has a cardinality of 220 degrees, diverse enough to perform information analysis.
- rating has low cardinality, showing that it is a set of classification values, used to classify data according to a certain condition.
- director has only completeness of 70.09%, nearly 30% of the data is missing, affecting the analysis process, making it easy to miss information.
- cast, country are also quite missing with 9-10% of the data.
- show_id (100% unique), title (99.99% unique, 1 duplicate), and description (99.64% unique, 32 duplicates) ensure the identification of each movie.
- After preprocessing the data, all NULL values have been processed, so the total is 0 completeness for all columns 100%.
- Actual (rows with values) is 8,807, equal to the total number of rows.
- country has 749 unique values (8.50%), showing the diversity of production countries. listed_in (genre) has 514 combinations (5.84%), reflecting movie titles in many different genres.
- type (2 values: Movie/TV Show), rating (18 values), release_year (74 values), and duration (221 values) have low uniqueness, suitable for classification roles, supporting trend analysis (e.g. by year, rating).

1.3 Profiling each attribute

Show id:

Metadata		show_id - Field Formats		
Attribute	Value	Format	Count	Percentage(%)
Field Name	show_id	XXXXX	7808	88.6568
Field Datatype	object	XXXX	900	10.2191
Field Length	1	XXX	90	1.0219
		XX	9	0.1022
Data Profiling Summary		show_id - Field Data Types		
Attribute	Value	Data Type	Count	Percentage(%)
NULL	0	object	8807	100
Missing	0			
Actual	8807			
Completeness (%)	100			
Cardinality	8807			
Uniqueness (%)	100			
Distinctness (%)	100			
Data Profiling Additional Statistics				
Attribute	Value			
Data Types	1			
Field Length (MIN)	2			
Field Length (MAX)	5			
Field Value (MAX)	N/A			
Field Formats	4			

General:

Completeness

- The show_id attribute has 100% completeness, no NULL values.
- The data is complete, no need to handle missing data.

Cardinality (Number of unique values)

- There are 8807 unique values, which are primary keys, each data is identified by an id.

Uniqueness & Distinctness

- Uniqueness: 100%
- Distinctness: 100%

Comments:

Distribution of show_id formats:

- show_id has a total of 4 main formats based on string length:
 - XXXXX – 7808 records (88.66%)
 - XXXX – 900 records (10.22%)
 - XXX – 90 records (1.02%)
 - XX – 9 records (0.10%)
- Although there is a difference in format, it does not affect the uniqueness of the value.

Data statistics:

- Length of show_id string:
 - Shortest: 2 characters
 - Longest: 5 characters
- Total number of different formats: 4
- Data type: object (string)

The show_id attribute acts as the primary key in the dataset.

type:

Metadata				type - Popular		
Attribute	Value			Value	Count	Percentage(%)
Field Name	type			Movie	6131	69.6151
Field Datatype	object			TV Show	2676	30.3849
Field Length	1					
Data Profiling Summary				type - Field Formats		
Attribute	Value			Format	Count	Percentage(%)
NULL	0			XXXXX	6131	69.6151
Missing	0			XXXXXXX	2676	30.3849
Actual	8807			type - Field Data Types		
Completeness (%)	100			Data Type	Count	Percentage(%)
Cardinality	2			object	8807	100
Uniqueness (%)	0.0227					
Distinctness (%)	0.0227					
Data Profiling Additional Statistics						
Attribute	Value					
Data Types	1					
Field Length (MIN)	5					
Field Length (MAX)	7					
Field Value (MAX)	N/A					
Field Formats	2					

General:

Completeness

- The type attribute has 100% completeness, no NULL values.
- The data is complete, no need to handle missing data.

Cardinality (Number of unique values)

- There are 2 unique values, corresponding to two types of content, Movie and TV Show.
- This shows that this is a binary categorical variable, easy to analyze and visualize.

Uniqueness & Distinctness

- Uniqueness: 0.0227% → Each value (Movie or TV Show) appears many times in the data set.
- Distinctness: 0.0227% → The data has a lot of repetition, because there are only 2 classification groups.

Remarks:

Distribution of content types

- The data about type is distributed as follows:
 - Movie: 6131 records – accounting for 69.62%.
 - TV Show: 2676 records – 30.38%.
- Movie is the most popular content type in the dataset.
- No outliers or errors → Reasonable distribution.

Data statistics

- Number of formats: 2 (XXXXXX and XXXXXX), consistent with the length of the two Movie and TV Show values.
- String length:
 - Longest: 7 characters (TV Show).
 - Shortest: 5 characters (Movie).

Categorical data

- All type data is object type (string).
- Data is clear, easy to process and suitable for classification analysis, visualization (pie chart, bar chart).

type is a high-quality, complete and clear classification attribute.

title:

Metadata		title - Field Formats			title - Field Data Types		
Attribute	Value	Format	Count	Percentage(%)	Data Type	Count	Percentage(%)
Field Name	title	XXXXXXXXXXXX	462	5.2458	object	8807	100
Field Datatype	object	XXXXXXXXXXXX	460	5.2231	title - Top 5 Most Popular		
Field Length	3	XXXXXXXXXXXX	452	5.1323	Value	Count	Percentage(%)
Data Profiling Summary		XXXXXXXXXXXX	450	5.1096	Consequences	2	0.0227
Attribute	Value	XXXXXXXXXXXX	436	4.9506	Ricardo O'Farrill Abrazo Genial	1	0.0114
NULL	0	XXXXXXXXXXXX	416	4.7235	Cooked	1	0.0114
Missing	0	XXXXXXXXXXXX	415	4.7122	Care Bears & Cousins	1	0.0114
Actual	8807	XXXXXXXXXXXX	410	4.6554	Hannibal Buress: Comedy Camisado	1	0.0114
Completeness (%)	100	XXXXXXXXXXXX	374	4.2466	title - Top 5 Least Popular		
Cardinality	8806	XXXXXXXXXXXX	369	4.1898	Value	Count	Percentage(%)
Uniqueness (%)	99.9886	XXXXXX	316	3.5881	ROAD TO ROMA	1	0.0114
Distinctness (%)	99.9886	XXXXXX	311	3.5313	Good Time	1	0.0114
Data Profiling Additional Statistics		XXXXXXXXXXXX	283	3.2134	Captain Underpants Epic Choice-o-Rama	1	0.0114
Attribute	Value	XXXXXXXXXXXX	276	3.1339	We Bare Bears	1	0.0114
Data Types	1	XXXXXXXXXXXX	256	2.9068	Zubaan	1	0.0114
Field Length (MIN)	1	XXXXXX	234	2.657			
Field Length (MAX)	104	XXXXXXXXXXXX	221	2.5094			
Field Value (MAX)	N/A	XXXXXXXXXXXX	212	2.4072			
Field Formats	77	XXXXXXXXXXXX	176	1.9984			
		XXXXXXXXXXXX	173	1.9643			
		XXXXXXXXXXXX	161	1.8281			
		XXXXXXXXXXXX	161	1.8281			
		XXXX	157	1.7827			
		XXXXXXXXXXXX	146	1.6578			
		XXXXXXXXXXXX	132	1.4988			
		XXXXXXXXXXXX	113	1.2831			
		XXXXXXXXXXXX	110	1.249			
		XXXXXXXXXXXX	108	1.2263			
		XXXXXXXXXXXX	92	1.0446			
		XXXXXXXXXXXX	87	0.9879			
		XXXXXXXXXXXX	81	0.9197			
		XXXXXXXXXXXX	76	0.8629			
		XXXXXXXXXXXX	58	0.6586			

General:

Completeness

- The title attribute is 100% complete, with no NULL values.
- The data is complete, no need to process missing data.

Cardinality

- There are 8806 unique values out of 8807 records, which shows that most of the titles are unique.
- There is only 1 duplicate value (a remake of a movie), which is a very small percentage.

Uniqueness & Distinctness

- Uniqueness: 99.9886%
- Distinctness: 99.9886%

→ Almost every title is unique, ensuring high diversity and distinction between records.

Remarks

Text data (Object)

- All data in the title is object type (string).
- No NULL values → Preprocessed.
- No duplicate values are visible as the titles are completely different except for remakes.

director:

Metadata		director - Field Data Types		
Attribute	Value	Data Type	Count	Percentage(%)
Field Name	director	object	8807	100
Field Datatype	object			
Field Length	3	director - Top 5 Most Popular		
Data Profiling Summary		Value	Count	Percentage(%)
Attribute	Value	Không có thông tin	2634	29.908
NULL	0	Rajiv Chilaka	19	0.2157
Missing	0	Raúl Campos, Jan Suter	18	0.2044
Actual	8807	Suhas Kadav	16	0.1817
Completeness (%)	100	Marcus Raboy	16	0.1817
Cardinality	4529	director - Top 5 Least Popular		
Uniqueness (%)	51.425	Value	Count	Percentage(%)
Distinctness (%)	51.425	Raymie Muzquiz, Stu Livingston	1	0.0114
Data Profiling Additional Statistics		Joe Menendez	1	0.0114
Attribute	Value	Eric Bross	1	0.0114
Data Types	1	Will Eisenberg	1	0.0114
Field Length (MIN)	2	Mozez Singh	1	0.0114
Field Length (MAX)	208			
Field Value (MAX)	N/A			
Field Formats	80			

General

Completeness

- The director attribute has a completeness of 100%, with no NULL or missing values.
- More than 29.9% of the records have a value of “No information”, which is equivalent to 2634 data that have been preprocessed to replace missing values.

Cardinality

- There are 4529 unique values out of a total of 8807 records.
- The level of distinction is relatively high, indicating that the data is rich in director identities and that many directors have made many different films.

Uniqueness & Distinctness

- Uniqueness: 51.425%
- Distinctness: 51.425% → The data has significant repetition between cholesterol levels.

→ More than half of the directors are unique, the rest may be directors of many different films or TV shows.

Remarks

Director Distribution

- There are more than 4500 different directors, showing a wide variety of data.
- The most common directors are:
 - Rajiv Chilaka (19 times)
 - Raúl Campos, Jan Suter (18 times)
 - Suhas Kadav, Marcus Raboy (16 times each)
- Many directors appear only once (e.g. Joe Menendez, Will Eisenberg, Mozez Singh,...).
- In addition, 29.9% of films and series do not have information about the director.
- This reflects the high level of dispersion of the director data field.

Text data (Object)

- All data in director are objects

- No NULL values → have gone through preprocessing.

Director is rich in information, suitable for analysis such as:

- Find top directors by number of films
- Compare directorial styles
- Analyze directorial impact on genre/popularity

Cast

Metadata		cast - Field Data Types			
Attribute	Value	Data Type		Count	Percentage(%)
Field Name	cast	object		8807	100
Field Datatype	object	cast - Top 5 Most Popular			
Field Length	3	Value		Count	Percentage(%)
Data Profiling Summary		Không có thông tin		825	9.3675
		David Attenborough		19	0.2157
		Vatsal Dubey, Julie Tejwani, Rupa Bhimani, Jigna Bhardwaj, Rajesh Kava, Mousam, Swapnil		14	0.159
		Samuel West		10	0.1135
		Jeff Dunham		7	0.0795
Attribute	Value				
NULL	0				
Missing	0				
Actual	8807				
Completeness (%)	100				
Cardinality	7693				
Uniqueness (%)	87.351				
Distinctness (%)	87.351				
Profiling Additional Statistics					
		Attribute	Value		
		Data Types	1		
		Field Length (MIN)	3		
		Field Length (MAX)	771		
		Field Value (MAX)	N/A		
		Field Formats	386		

General:

Completeness

- High Completeness (100%) with no NULL values.

- However, there are 825 records (9.37%) with the value "No information" → a high proportion will affect the analysis, visualization and comparison process.

Cardinality (Number of unique values)

- There are 7693 unique values out of a total of 8807 records → showing a very high diversity and richness in actor data, with changes in the number of actors in movies.

Uniqueness & Distinctness

- Uniqueness: 87.351%
- Distinctness: 87.351%

→ Most records have different casts, only a few shows/movies have the same cast.

Comments:

Text data (Object)

- Data type: object
- Field length:

- Minimum: 3 characters
- Maximum: 771 characters → can contain long lists of actors (names separated by commas or otherwise).

Number of different formats: 386 → shows that the actor name format is not uniform, may need to be standardized (especially if separating individual actor names for network or frequency analysis).

→ It is easy to see that no actor appears too many times, actor data is strongly dispersed → consistent with the characteristics of diverse movies and TV shows.

Country

Metadata		country - Field Formats			country - Field Data Types		
Attribute	Value	Format	Count	Percentage(%)	Data Type	Count	Percentage(%)
Field Name	country	XXXXXXXXXXXXXXXXXX	2862	32.4969	object	8807	100
Field Datatype	object	XXXXX	1599	18.156	country - Top 5 Most Popular		
Field Length	3	XXXXXXXXXXXXXXXXXX	1276	14.4885	Value	Count	Percentage(%)
Data Profiling Summary		XXXXXX	763	8.6636	United States	2818	31.9973
Attribute	Value	XXXXXXXXXXXXXXXXXX	310	3.5199	India	972	11.0367
NULL	0	XXXXXXX	301	3.4177	Không xác định	831	9.4357
Missing	0	XXXXXXXXXXXXXXXXXX	248	2.8159	United Kingdom	419	4.7576
Actual	8807	XXXXXXXXXXXXXXXXXXXXXXXXXXXX	228	2.5888	Japan	245	2.7819
Completeness (%)	100	XXXXXXXXXXXXXXXXXXXXXXXXXXXX	157	1.7827	country - Top 5 Least Popular		
Cardinality	749	XXXXXXXXXX	140	1.5896	Value	Count	Percentage(%)
Uniqueness (%)	8.5046	XXXXXXXXXXXXXXXXXXXXXXXXXXXX	102	1.1582	Romania, Bulgaria, Hungary	1	0.0114
Distinctness (%)	8.5046	XXXXXXXXXXXXXXXXXXXXXXXXXXXX	78	0.8857	Uruguay, Guatemala	1	0.0114
Data Profiling Additional Statistics		XXXXXXXXXXXXXXXXXXXXXXXXXXXX	55	0.6245	France, Senegal, Belgium	1	0.0114
Attribute	Value	XXXXXXXXXXXXXXXXXXXX	54	0.6131	Mexico, United States, Spain, Colombia	1	0.0114
Data Types	1	XXXXXXXXXXXXXXXXXXXX	53	0.6018	United Arab Emirates, Jordan	1	0.0114
Field Length (MIN)	4	XXXXXXXXXXXXXXXXXXXX	49	0.5564			
Field Length (MAX)	123	XXXXXXXXXXXXXXXXXXXX	47	0.5337			
Field Value (MAX)	N/A	XXXXXXXXXXXXXXXXXXXXXXXXXXXX	42	0.4769			
Field Formats	72	XXXXXXXXXXXXXXXXXXXXXXXXXXXX	34	0.3861			
		XXXXXXXXXXXXXXXXXXXXXXXXXXXX	32	0.3633			
		XXXXXXXXXXXXXXXXXXXXXXXXXXXX	28	0.3179			
		XXXXXXXXXXXXXXXXXXXXXXXXXXXX	28	0.3179			
		XXXXXXXXXXXXXXXXXXXXXXXXXXXX	28	0.3179			
		XXXXXXXXXXXXXXXXXXXXXXXXXXXX	27	0.3066			
		XXXXXXXXXXXXXXXXXXXXXXXXXXXX	17	0.193			
		XXXXXXXXXXXXXXXXXXXXXXXXXXXX	15	0.1703			
		XXXXXXXXXXXXXXXXXXXXXXXXXXXX	15	0.1703			
		XXXXXXXXXXXXXXXXXXXXXXXXXXXX	15	0.1703			
		XXXXXXXXXXXXXXXXXXXX	14	0.159			
		XXXXXXXXXXXXXXXXXXXXXXXXXXXX	14	0.159			
		XXXXXXXXXXXXXXXXXXXXXXXXXXXX	13	0.1476			
		XXXXXXXXXXXXXXXXXXXXXXXXXXXX	12	0.1363			
		XXXXXXXXXXXXXXXXXXXX	12	0.1363			
		XXXXXXXXXXXXXXXXXXXX	11	0.1249			
		XXXXXXXXXXXXXXXXXXXX	11	0.1249			
		XXXXXXXXXXXXXXXXXXXXXXXXXXXX	9	0.1022			
		XXXXXXXXXXXXXXXXXXXXXXXXXXXX	8	0.0908			
		XXXXXXXXXXXXXXXXXXXXXXXXXXXX	8	0.0908			
		XXXXXXXXXXXXXXXXXXXX	7	0.0795			
		XXXXXXXXXXXXXXXXXXXXXXXXXXXX	6	0.0681			

General:

Completeness

- High completeness (100%) with no NULL values
- However, there are 831 records (9.4357%) with the value "Unknown".

Cardinality

- Cardinality: 749 → many different countries or country combinations (e.g. “France, Spain”, “Mexico, United States...”)
- Uniqueness/Distinctness: 8.5046% → low repetition, very dispersed data, showing richness in production origins.

Remarks:

- The United States accounts for nearly 32%, followed by India (~11%) and the United Kingdom (~5%).
- There are many records containing multiple countries: e.g. France, Spain, Germany → this shows internationality and multinational cooperation in content production.
- The unknown data is relatively large, almost 1/10 of the entire set → this directly affects geographically related analyses.

Date added

Metadata		date_added - Top 5 Least Popular		
Attribute	Value	Value	Count	Percentage(%)
Field Name	date_added	December 4, 2016	1	0.0114
Field Datatype	object	November 21, 2016	1	0.0114
Field Length	2	November 19, 2016	1	0.0114
Data Profiling Summary		November 17, 2016	1	0.0114
		January 11, 2020	1	0.0114
		date_added - Field Formats		
		Format	Count	Percentage(%)
		XXXXXXXXXXXXXXXXXX	1589	18.0425
Attribute	Value	XXXXXXXXXXXXXXXXXX	1567	17.7927
NULL	0	XXXXXXXXXXXXXXXXXX	1462	16.6004
Missing	0	XXXXXXXXXXXXXXXXXX	1258	14.2841
Actual	8807	XXXXXXXXXXXXXXXXXX	1178	13.3757
Completeness (%)	100	XXXXXXXXXXXXXX	1040	11.8088
Cardinality	1768	XXXXXXXXXXXXXXXXXX	439	4.9847
Uniqueness (%)	20.0749	XXXXXXXXXXXXXX	272	3.0885
Distinctness (%)	20.0749	XXXXXXXXXXXXXXXXXX	2	0.0227
Data Profiling Additional Statistics		date_added - Field Data Types		
Attribute	Value	Data Type	Count	Percentage(%)
Data Types	1	object	8807	100
Field Length (MIN)	11			
Field Length (MAX)	19			
Field Value (MAX)	N/A			
Field Formats	9			
date_added - Top 5 Most Popular				
Value	Count	Percentage(%)		
January 1, 2020	109	1.2377		
November 1, 2019	89	1.0106		
March 1, 2018	75	0.8516		
December 31, 2019	74	0.8402		
October 1, 2018	71	0.8062		

General:

Completeness

- High Completeness (100%) with no NULL values

Cardinality

- Cardinality: 1768, meaning there are 1768 different date values in 8807 records.
- Uniqueness/Distinctness: 20.07% → moderate duplication, indicating some days have multiple contents added at the same time (suitable for batch updates).

Remarks:

Text data (Object)

Data type: object

String length:

- Minimum: 11 characters
- Maximum: 19 characters

→ indicates some formats including time or month name length (e.g. "September 15, 2019").

Completeness: 100% → no missing values (NULL or Missing).

Cardinality: 1768 → there are 1768 different dates out of 8807 records.

Uniqueness/Distinctness: ~20.07% → relatively high level of duplication, indicating that many records were added on the same day → reflecting batch uploads.

Most popular date: January 1, 2020 (1.24%)

The date_added column indicates when the movie was uploaded to Netflix.

release_year

Metadata		release_year - Top 5 Most Popular		
Attribute	Value	Value	Count	Percentage(%)
Field Name	release_year	2018	1147	13.0237
Field Datatype	int64	2017	1032	11.718
Field Length	1	2019	1030	11.6952
		2020	953	10.8209
		2016	902	10.2419
Data Profiling Summary		release_year - Top 5 Least Popular		
Attribute	Value	Value	Count	Percentage(%)
NULL	0	1959	1	0.0114
Missing	0	1925	1	0.0114
Actual	8807	1961	1	0.0114
Completeness (%)	100	1947	1	0.0114
Cardinality	74	1966	1	0.0114
Uniqueness (%)	0.8402			
Distinctness (%)	0.8402			
Data Profiling Additional Statistics		release_year - Field Formats		
Attribute	Value	Format	Count	Percentage(%)
Data Types	1	XXX	8807	100
Field Length (MIN)	4			
Field Length (MAX)	4	release_year - Field Data Types		
Field Value (MAX)	2021	Data Type	Count	Percentage(%)
Field Formats	1	int64	8807	100

General:

- High Completeness (100%) with no NULL values.
- There are 74 unique values, corresponding to 74 different years (from to 2021) allowing for analysis of film release trends over time.
- Uniqueness / Distinctness: 0.8402% → most years have many repeated records. For example, 2018 alone had 1,147 records.

Remarks:

- More than 57% of the content comes from the 2016–2020 period → showing a platform trend of prioritizing new, modern content that keeps up with the market.
- Some old years (like 1959, 1961) have only 1 record → representing classic or rare content that is rarely distributed or updated.

rating:

Metadata				rating - Top 5 Least Popular		
Attribute	Value			Value	Count	Percentage(%)
Field Name	rating			NC-17	3	0.0341
Field Datatype	object			UR	3	0.0341
Field Length	2			74 min	1	0.0114
				84 min	1	0.0114
Data Profiling Summary				66 min	1	0.0114
Attribute	Value					
NULL	0			rating - Field Formats		
Missing	0			Format	Count	Percentage(%)
Actual	8807			XXXXX	7057	80.1294
Completeness (%)	100			X	840	9.5379
Cardinality	18			XXXX	527	5.9839
Uniqueness (%)	0.2044			XX	370	4.2012
Distinctness (%)	0.2044			XXXXXXXXXX	6	0.0681
				XXXXXXXXXXXXXXXXXX	4	0.0454
Data Profiling Additional Statistics				XXXXXX	3	0.0341
Attribute	Value					
Data Types	1			rating - Field Data Types		
Field Length (MIN)	1			Data Type	Count	Percentage(%)
Field Length (MAX)	14			object	8807	100
Field Value (MAX)	N/A					
Field Formats	7					
rating - Top 5 Most Popular						
Value	Count	Percentage(%)				
TV-MA	3207	36.4142				
TV-14	2160	24.5259				
TV-PG	863	9.799				
R	799	9.0723				
PG-13	490	5.5638				

General:

- Completeness is 100%.
- There are 18 unique values → representing many age classification levels, from children to adults, the diversity of content types, serving many different target groups.
- Uniqueness / Distinctness index: 0.2044% → each rating value appears many times in the dataset, because there are only 18 values.

Detailed comments:.

- The content mainly focuses on TV-MA and TV-14 classifications, targeting adult and teen audiences - this is the main user file of platforms like Netflix.
- NC-17, UR, 74 min, 84 min, 66 min appear only 1–3 times → accounting for < 0.04% each.

Duration

Metadata				duration - Top 5 Most Popular		
Attribute	Value			Value	Count	Percentage(%)
Field Name	duration			1 Season	1793	20.3588
Field Datatype	object			2 Seasons	425	4.8257
Field Length	2			3 Seasons	199	2.2596
				90 min	152	1.7259
Data Profiling Summary				94 min	146	1.6578
Attribute	Value					
NULL	0			duration - Top 5 Least Popular		
Missing	0			Value	Count	Percentage(%)
Actual	8807			189 min	1	0.0114
Completeness (%)	100			10 min	1	0.0114
Cardinality	221			3 min	1	0.0114
Uniqueness (%)	2.5094			229 min	1	0.0114
Distinctness (%)	2.5094			191 min	1	0.0114
Data Profiling Additional Statistics				duration - Field Formats		
Attribute	Value			Format	Count	Percentage(%)
Data Types	1			XXXXXX	3202	36.3574
Field Length (MIN)	5			XXXXXXXX	2922	33.1782
Field Length (MAX)	14			XXXXXXXXXX	1793	20.3588
Field Value (MAX)	N/A			XXXXXXXXXXX	866	9.8331
Field Formats	7			XXXXXXXXXXXX	17	0.193
				XXXXX	4	0.0454
				XXXXXXXXXXXXXXXX	3	0.0341
				duration - Field Data Types		
				Data Type	Count	Percentage(%)
				object	8807	100

General:

- 100% complete data, no NULL or missing values.
- There are 221 unique values, clearly showing the diversity of content duration (including both individual movie duration and number of TV Show seasons).
- Uniqueness / Distinctness: 2.5094% → each value is repeated many times, almost the usual duration of movies of each genre is like .

Detailed comments:

- TV Shows account for a large proportion of seasons (especially 1 Season), showing that episodic content is very popular. → Meanwhile, the most popular individual movies have durations of 90 and 94 minutes, consistent with the average length of movies.
- Durations such as 189 min, 10 min, 3 min, 229 min, 191 min appear only once..

Discription

Metadata		description - Field Data Types		
Attribute	Value	Data Type	Count	Percentage(%)
Field Name	description	object	8807	100
Field Datatype	object			
Field Length	3	description - Top 5 Most Popular		
Data Profiling Summary		Value	Count	Percentage(%)
Attribute	Value	Paranormal activity at a lush, abandoned property alarms a group eager to redevelop the site, but the eerie events may not be as unearthly as they think.	4	0.0454
NULL	0	Challenged to compose 100 songs before he can marry the girl he loves, a tortured but passionate singer-songwriter embarks on a poignant musical journey.	3	0.0341
Missing	0	A surly septuagenarian gets another chance at her 20s after having her photo snapped at a studio that magically takes 50 years off her life.	3	0.0341
Actual	8807	Multiple women report their husbands as missing but when it appears they are looking for the same man, a police officer traces their cryptic connection.	3	0.0341
Completeness (%)	100	Secrets bubble to the surface after a sensual encounter and an unforeseen crime entangle two friends and a woman caught between them.	2	0.0227
Cardinality	8775			
Uniqueness (%)	99.6367			
Distinctness (%)	99.6367			
Data Profiling Additional Statistics				
Attribute	Value			
Data Types	1			
Field Length (MIN)	61			
Field Length (MAX)	248			
Field Value (MAX)	N/A			
Field Formats	104			

Overview:

- 100% complete data, no NULL or missing values.
- Uniqueness & Distinctness both reach 99.6367%.
- Duplicate descriptions can be a re-released movie or a special version of a movie.

Detailed comments:

- Shortest description: 21 characters
- Longest description: 248 characters
- A total of 104 descriptions of different lengths → the movie content is very diverse and non-duplicated, suitable for all genres.

2. Abstraction

2.1. Domain task 1: Analyze content trends by year and type.

2.1.1. Data abstraction

Data:	netflix_titles_cleaned						
Item:	1 dòng là 1 nội dung (Movie hoặc TV Show)						
Dataset availability:	Static						
	Semantics	Attribute Type	Hierarchical	Characteristic	Direction	Quantitative Type	Bin Number
Release Year	Năm phát hành	Quantitative	None	Discrete	Sequential	Interval	13 (2008–2020)
Type	Loại nội dung	Categorical	None	Discrete	None	None	2 (Movie, TV)

2.1.2. Task abstraction

Produce → Explore → Sum

- **Produce**

- o Check for missing values in the Release Year and Type columns..
- o Remove rows without a release year..
- o Combine content counts by Release Year and Type.

- **Explore**

Exploratory Questions:

- o Is there a difference between the number of Movies and TV Shows over the years?
- o Which year had the biggest growth?

Act::

- o Count the number of content per year (Release Year) divided by type (Movie, TV Show).
- o Draw an Area Chart to show changes over time::
 - Trục X: Release Year
 - Trục Y: Count of Show ID
 - Color distinguishes between Movies and TV Shows

- **Sum**

Comment::

- o Since 2016, the total amount of Netflix content has grown significantly.
- o Movies always dominate, but TV Shows tend to increase sharply after 2017.

Summary of findings:

- o Significant growth: 2016–2019 period.
- o Movies make up the bulk of the offering, but TV Shows play an increasingly important role in Netflix's content strategy.

2.2. Domain task 2: Analyzing the countries that produce the most content

2.2.1. Data abstraction

Data:	netflix_titles_cleaned						
Item:	1 dòng là 1 nội dung (Movie hoặc TV Show)						
Dataset availability:	Static						
Attribute	Semantics	Attribute Type	Hierarchical	Characteristic	Direction	Quantitative Type	Bin Number
Country	Quốc gia sản xuất	Categorical	None	Discrete	None	None	~100+
Show ID	ID nội dung	Identifier	None	Discrete	None	Count-based	–

2.2.2. Task abstraction

Produce → Explore → Sum

• Produce

- o Remove missing or "Unknown" values in the Country column.
- o Separate entries with multiple countries (if necessary) — for example "India, United States" → split into 2 lines (if you choose to be more specific).
- o Count the amount of content produced in each country.

• Explore

Exploratory Questions:

- o Which country has the most Netflix content?
- o Is content concentrated in a few countries or is it dispersed?

Act:

- o Create a horizontal Bar chart (Horizontal Bar Chart):
 - Y-axis: Country

- X-axis: Count of Show ID
- Sort by quantity in descending order
- Select to display Top 10 countries

- **Sum**

Comment::

- o The United States is the country with the most content — over 2,000 entries.
- o India and some countries like UK, Canada, Japan are also important content production centers.

Summary of findings:

- o Netflix tends to focus content from a few key countries, primarily English speaking.
- o This reflects a strategy of globalizing content but still prioritizing markets with large entertainment output.

2.3. Domain task 3: Movie genre trends have been updated a lot in recent years.

2.3.1. Data abstraction

Attr	Abstraction							
Data Type	Attribute and Item							
Dataset type	Table							
Dataset availability	Static							
No	Attribute Name	Type(C, Q, Q)	Interval/ Ratio	Key/ Value	Direction	Hierarchical	Continuous/ Discrete	Semantic
1	Date Added	Quantitative	Interval	Value	Sequential	Yes	Discrete	Yes. Đây là ngày nội dung được thêm vào Netflix, dùng để phân tích xu hướng thể loại theo quý. "Yes, nội dung được thêm vào quý nào?"
2	Listed In	Categorical	None	Key	None	No	Discrete	Yes. Đây là danh sách thể loại của nội dung, dùng để xác định xu hướng thể loại phổ biến. "Yes, thể loại nào đang thịnh hành?"

2.3.2. Task abstraction

Analysis **Analyze** → **Search** → **Query**

- **Analyze:**
 - **Consume:** ate added (date_added) and genre (listed_in) data for Netflix movies and shows is now available.
 - **Perform:** Can use statistical tables or charts (e.g., line charts to show genre trends over quarters, or bar charts to compare genre proportions in a particular quarter) → **Present**
- **Search:**
 - No need to search for a specific movie or show, but want to observe the general trend of genres over the quarters.
 - Not suitable for Lookup or Locate because the domain task does not aim to find a

specific content but focuses on discovering the relationship between time (date_added) and category (listed_in) → Explore.

- **Query:**
 - Find trends in categories over quarters (e.g. which categories are increasing or decreasing over time) and sum up the number or percentage of categories by quarter → Summarize
- ⇒ **Present** → **Explore** → **Summarize**

2.4. Domain task 4: Analyzing the Relationship Between Age Rating and Content Genre

2.4.1. Data abstraction

Attr	Abstraction							
Data Type	Attribute and Item							
Dataset type	Table							
Dataset availability	Static							
No	Attribute Name	Type(C, O, Q)	Interval/ Ratio	Key/ Value	Direction	Hierarchical	Continuuous/ Discrete	Semantic
1	Rating	Categorical	None	Key	None	Yes	Discrete	Yes. Đây là phân loại độ tuổi của nội dung, dùng để phân tích mối quan hệ với thể loại nội dung. "Yes, phân loại độ tuổi này phổ biến với thể loại nào?"
2	Listed In	Categorical	None	Key	None	No	Discrete	Yes. Đây là danh sách thể loại của nội dung, dùng để xác định mối quan hệ với phân loại độ tuổi. "Yes, thể loại này thường có phân loại độ tuổi nào?"

2.4.2. Task abstraction

Analysis by **Analyze** → **Search** → **Query**

Analyze:

- **Consume:** Data on the age ratings (ratings) and genres (listed_in) of Netflix content is available in the file netflix_titles_cleaned.csv.
- **Search:** A visual table or graph can be used to present the relationship between age rating and content type. A Treemap can display the number of content items for each rating and listed_in pair, with the rectangle size and color indicating the number → **Present**

Search:

- Don't look for a specific movie or show, but focus on observing the overall relationship between age rating and content genre.
- Not suitable for Lookup or Locate because the domain task "Analyze the relationship between Age Rating and Content Genre" is not intended to search for a specific content (like looking up a movie) or locate an item (like finding the location of a show). Instead, it analyzes the relationship between rating and listed_in, for example, see what genre "TV-MA" is often associated with → **Explore**

Query:

- Find the relationship between age classification and content genre (for example, "TV-Y" content is often included in "Kids' TV", or "TV-MA" is often associated with "TV Dramas" and "International TV Shows") and summarize the number or proportion of genres by age classification to draw conclusions → **Summarize**

⇒ **Present** → **Explore** → **Summarize**

2.5. Domain task 5: Comparing the amount of content across audience age groups

2.5.1. Data abstraction

Attr	Abstraction							
Data type	Attribute and Item							
Dataset type	Table							
Dataset availability	Static							
No	Attribute Name	Type(C, O, Q)	Interval/Ratio	Key/Value	Direction	Hierarchical	Continuous/Discrete	Semantic
1	Rating	Categorical	None	Value	Sequential	Yes	Discrete	Yes. Đại cho độ tuổi khán giả mục tiêu và mức độ hạn chế nội dung
2	Number of records	Quantitative	None	Key	None	No	Discrete	No

2.5.2. Task abstraction

Process analysis Analyze → Search → Query

Analyze:

- **Consume:**
 - Data on the age ratings and number of records of movies and shows on Netflix is available.
 - Details::
 - Ratings are classification values (G, TV-G, TV-Y, ..., TV-MA, NC-17, NR, UR), representing the age of the target audience and the level of content restrictions.
 - Number of records is the number of contents corresponding to each age level, which has been compiled in advance.
- **Search:**
 - You can use a column chart to show the number of records by age rating.
 - In the current chart, Ratings are sorted in natural order (G → UR) to highlight trends by audience age.

→ **Present:** The data is ready to be presented directly as a column chart.

Search:

- **No need to search for a specific age range:**
 - The chart does not focus on finding a specific norm (e.g. just looking at the amount of TV-MA content), but rather looks at the general trend and distribution across all age norms (G → UR).
- **Not suitable for Lookup or Locate:**
 - Domain tasks do not aim to find a specific value (Lookup, e.g. the number of contents of TV-MA) or determine the location of a value (Locate, e.g. where TV-MA is in the list).
 - Instead, the domain task focuses on exploring the relationship between the number of records and the age of the target audience (represented by the Rating order).

→ **Explore:** The task is to discover trends and distributions across the entire data.

Query:

- **Find trends and distributions:**
 - Domain task requires finding trends in the amount of content across audience age

(from children to adults) and aggregating the amount of content by age category.

○ Example:

- How does the amount of content increase or decrease as audience age increases (G → UR)?
- Which age rating has the largest amount of content?

→ **Summarize**: Summarize trends and distributions across the entire data.

=> **Present** → **Explore** → **Summarize**.

2.6. Domain task 6: Compare content quantity between producing countries

2.6.1. Data abstraction

	Latitude (generated)	Quantitative	Interval	Value	Sequential	No	Continuous	Yes. Kinh độ địa lý quốc gia
4	Longitude (generated)	Quantitative	Interval	Value	Sequential	No	Continuous	Yes. Kinh độ địa lý quốc gia

2.6.2. Task abstraction

Process analysis Analyze → Search → Query

Analyze

Consume:

- **Data on country of production (Country), content quantity (CNT), and geographic location (Latitude, Longitude):** Data is available from the netflix_titles_cleaned.csv dataset.
 - **Country:** A categorical value (e.g. United States, India, South Korea), representing the country where the content was produced.
 - **CNT (Number of records):** The number of records corresponding to each country, pre-aggregated (e.g. United States: 2,818, India: 912).
 - **Latitude and Longitude:** Fields automatically created in Tableau to locate countries on the map.

Detail:

- **Country** is a classification value that distinguishes countries that produce content.
- **CNT** is a quantitative value that represents the amount of content produced by each country.
- **Latitude và Longitude** is an ordinal value, used to map the geographic location of countries on a map.

Perform:

- The current map uses a **map chart (Map)** to represent the content quantity (CNT) by country (Country), with countries located by Latitude and Longitude.
- **Encoding:** The content count (CNT) is color-coded (on a scale from light yellow to dark red), where darker colors indicate larger content counts (e.g., United States in dark red has 2,818 titles).
- **Focus:** The chart focuses on geographical distribution and compares the amount of content across countries.

→ **Present:** The data is ready to be presented directly as a map with color coding of the amount of content.

Search

No need to search for a specific country or region:

- The map does not focus on finding the amount of content for a specific country (e.g., only see United States) or a specific region (only see North America), but rather looks at the general distribution across all countries and geographic regions.
- The domain task aims to compare the amount of content across countries and discover geographic areas with large content production.

Not suitable for Lookup or Locate:

- Domain tasks are not aimed at finding a specific value (Lookup, e.g., number of United States content) or determining the location of a country on a map (Locate, e.g., where is United States located).
- Instead, the task focuses on exploring the relationship between content quantity (CNT) and geographic location (Country), and comparing across countries and regions.

→ **Explore:** The task is to explore the distribution and make comparisons across the entire data.

Query

Find the distribution and compare:

- Domain task requires finding the distribution of content volume by country and comparing it across geographic regions (e.g. North America, South Asia, East Asia).
- For example:
 - Which region produces the most content (e.g. North America vs. United States vs. Canada)?
 - What is the difference in the amount of content between large countries like the United States and India compared to smaller countries like Guatemala or Cambodia?

→ **Summarize:** Summarize distribution and comparison across entire data

=> **Present** → **Explore** → **Summarize.**

2.7. Domain task 7: Actor distribution by genre

2.7.1. Data abstraction

Attr	Abstraction							
Data type	Attribute and Item							
Dataset type	Table							
Dataset availability	Static							
No	Attribute Name	Type(C,O,Q)	Interval/Ratio	Key/Value	Direction	Hierarchical	Continuous/Discrete	Semantic
1	Cast	Quantitative	Ratio	Value	None	No	Discrete	Yes. Số lượng diễn viên
2	Listed in	Categorical	None	Key	None	Yes(Genres)	Discrete	Yes. Danh sách thể loại

2.7.2. Data abstraction

Analyze -> Search -> Query

Analyze:

- **Consume:**

- Data on number of actors (Cast), genre (Listed in).
- Detail:
 - Listed in are the category-distinguishing values..
 - Number of actors (cast).

→ **Present**

Search:

- **No need to search for specific genres:**

- The chart does not focus on finding the number of actors of a particular genre but rather looks at the distribution of the number of actors across genres.

- **Not suitable for Lookup or Locate:**

- Domain tasks do not aim to find a specific value or determine the location of a value (Lookup or Locate).
- Instead, the domain task focuses on discovering the distribution of actor counts by movie genre.

→ **Explore:** The task is to explore the distribution and make comparisons across the entire data.

Query:

- **Find distribution and compare:**

- Domain task requires finding the distribution of actors by genre so that they can be compared and evaluated.
- Example: Based on the visualization, which genre has the most and least number of actors?

→ **Summarize:** Summarize data distribution and compare across entire dataset.

⇒ **Present → Explore → Summarize**

2.8. Domain task 8: Distribution and trends of age classification by year of release

2.8.1. Data abstraction

Attr	Abstraction							
Data type	Attribute and Item							
Dataset type	Table							
Dataset availability	Static							
No	Attribute Name	Type(C,O,Q)	Interval/Ratio	Key/Value	Direction	Hierarchial	Continuos/Discrete	Semantic
1	Rating	Categorical	None	Key	None	Yes(Rating)	Discrete	Yes. Phân loại độ tuổi
2	Release year	Quantitative	Interval	Key	Increasing	Yes(Years)	Discrete	Yes. Năm phát hành

2.8.2. Task abstraction

Analyze:

- **Consume**

- Data on age rating and released year.
- Detail:
 - Release year The release year of movies with age ratings.
 - Number of age ratings.

→ Present

Search:

- **No need to search for specific genres::**

- The chart does not focus on finding specific age categories or release years, but rather focuses on the distribution and trends of age categories by release year.

- **Not suitable for Lookup or Locate:**

- Domain tasks do not aim to find a specific value or determine the location of a value (Lookup or Locate).
- Instead, the domain task focuses on discovering the distribution of age-rated movies by year of release.

→ Explore: The task is to discover distributions and trends across the entire data.

- **Query:**

- **Find distribution and compare:**

- Domain task requires finding the distribution of the number of movies classified by age by year of release.
- For example:
 - Based on visualization, can we show trends in age classifications over each year of release?

▪ Which year and genre has the highest and lowest number of releases by age group?
→ **Summarize**: Tóm tắt phân phối và xu hướng dữ .

=> **Present** → **Explore** → **Sum**

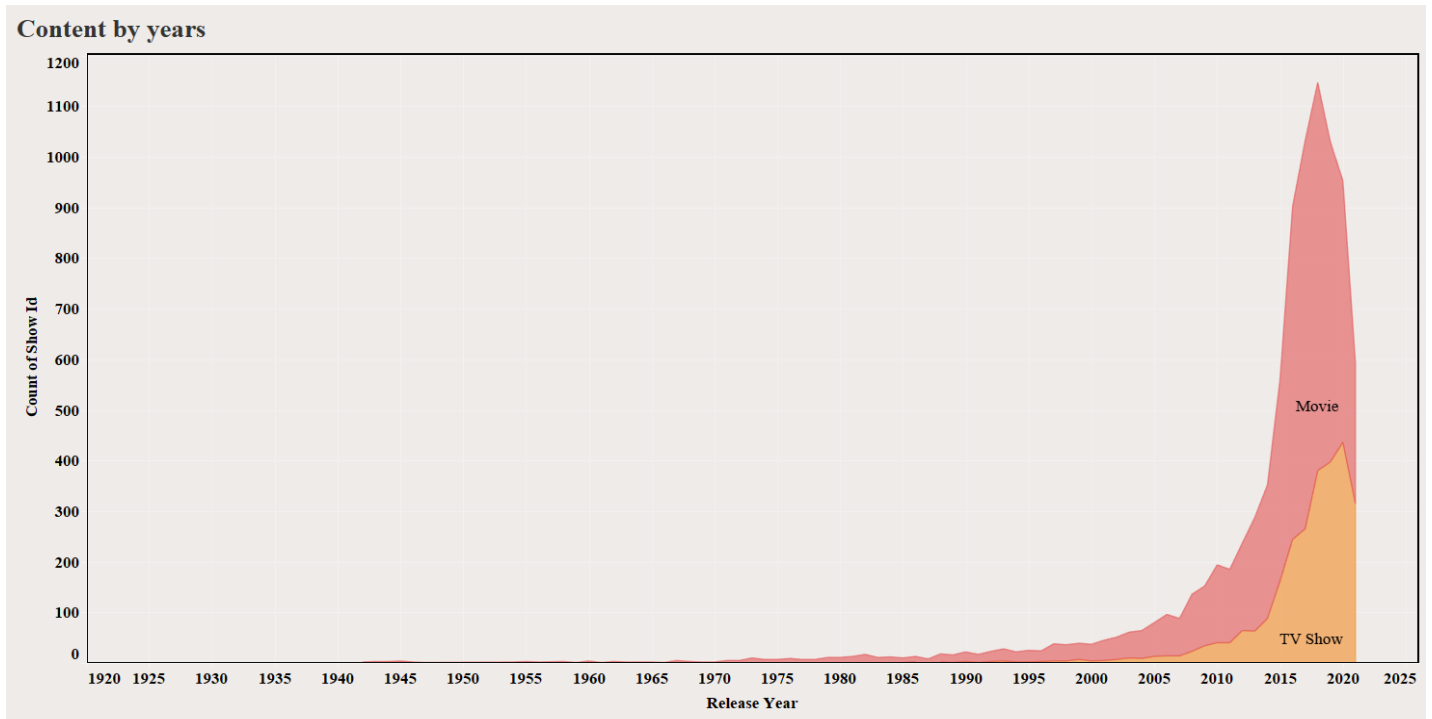
3. Idiom design

3.1.Domain task 1

3.1.1. Idiom

Idiom		Bar charts	
Data	Content Type: C (Categorical, TV Show)		
	Release Year: O (Ordinal - year of release)		
	Count of Show ID: Q (Quantitative - amount of content)		
Encode	Mark: area		
		O: horizontal position	
	Channel	Q: vertical position	
		C: color (distinguishing between Movie and TV Show)	
TASK	<i>Track the increase/decrease in content volume over time</i> <i>Comparison between Movie TV and TV Show over the years</i>		
SCALE	<i>Keys: ~13</i> <i>(2008-2020)</i> <i>Level:</i> <i>Ordinal</i>		

3.1.2. Chart



3.1.3. Evaluate

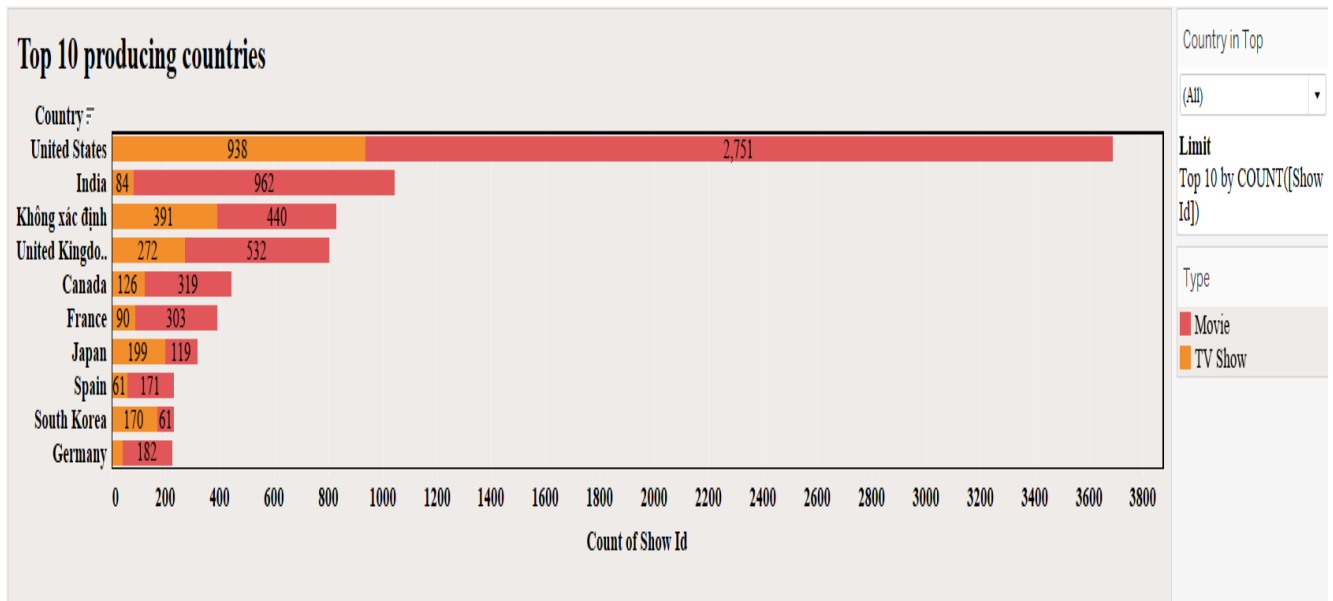
- **Expressiveness**
 - The chart shows the full release years and corresponding content amounts.
 - Clearly distinguish two types of content with prominent colors.
 - Area and slope show growth rate.
- **Efficiency**
 - Accuracy: The Y-axis represents quantity for accurate value reading.
 - Discriminability: Colors clearly distinguish between Movies and TV Shows.
 - Separability: Two types of content displayed on the same chart make it easy to compare percentages year by year.
- **Chart analysis**
 - Helps identify the age at highest risk of heart disease and trends by age.

3.2.Domain task 2

3.2.1. Idiom

Idiom	Pie charts	
Data	Country: O (Ordinal -> country name)	
	Content Type: C (Categorical – Movie, TV Show)	
	Count of Show ID: Q (Quantitative - amount of content)	
Encode	Mark: bar	
	Channel	O: vertical pos (country - Y axis)
		Q: horizontal pos (quantity - X axis)
		C: color (distinguishing between Movie and TV Show)
TASK	Compare content production levels across countries. Observe which countries dominate and what types of content are popular in each country.	
SCALE	Keys: 10 (Top 10 countries) Level: Ordinal (ranked by quantity)	

3.2.2. Chart



3.2.3. Evaluation

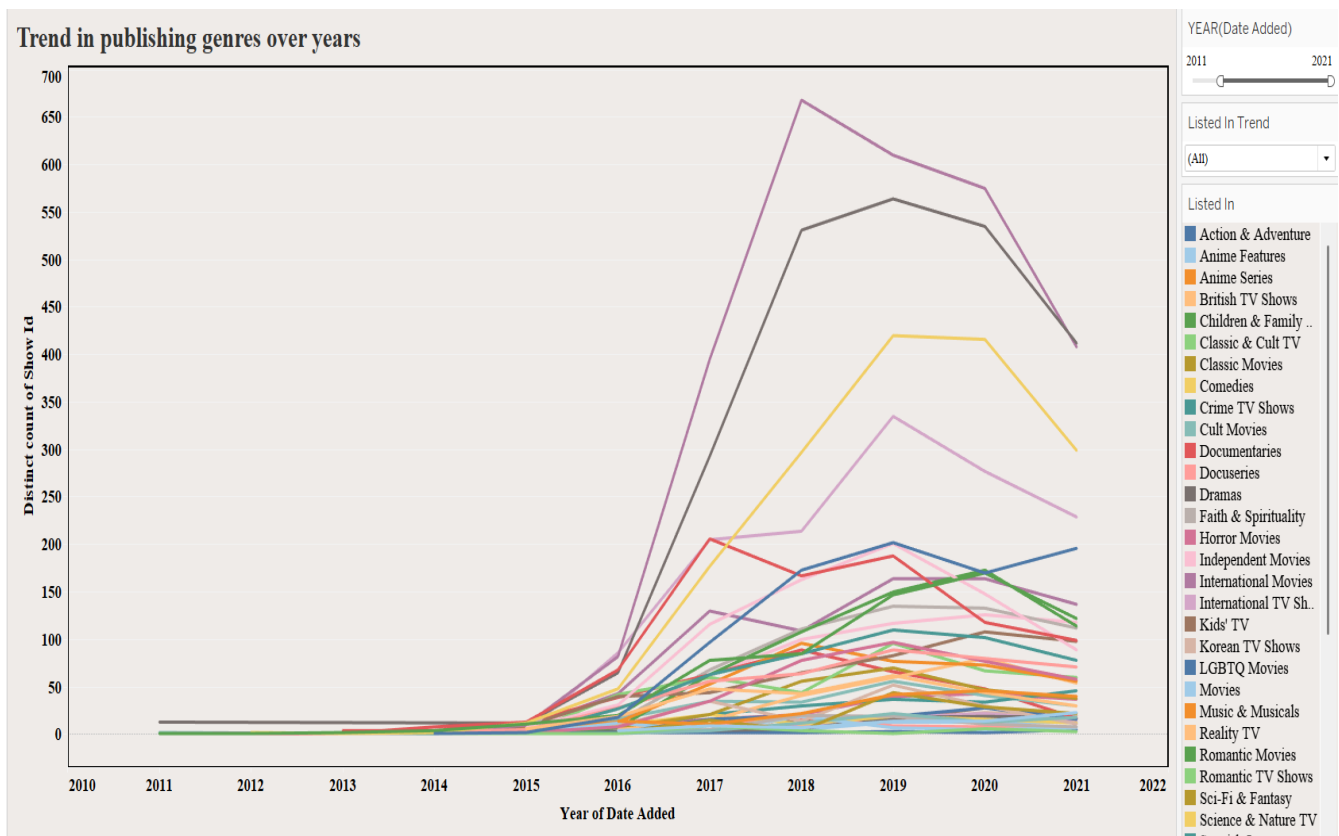
- **Expressiveness**
 - Circular chart helps to clearly show the proportion of each group.
 - Easily identify the difference between men and women.
- **Efficiency**
 - Accuracy: helps to easily visualize the ratio between groups.
 - Discriminability: using color to differentiate between having/not having heart disease.
 - Separability: not considered because there is no attribute that uses 2 channels to represent.
- **Chart analysis**
 - Helps determine which gender has a higher rate of heart disease and the extent of the difference between the two genders.

3.3.Domain task 3

3.3.1. Idiom

Idiom	Lines	
Data	Count: Q (Quantitative)	
	Listed In: C (Categorical)	
	Date Added: Q (Quantitative)	
Encode	Mark: area chart	
	Channel	Q: vertical pos. The number of items (Count) is encoded by the vertical position on the Y axis
		Q: horizontal pos. The time (date_added) is encoded by the horizontal position on the X axis.
		C: Color: Listed_in pairs (e.g. Kids' TV") are color coded.
TASK	Comparing the amount of content by genre uploaded to Netflix over the years. Genre trends uploaded to Netflix over the quarters.	
SCALE	Keys:10 Level: Categorical (Listed In), Quantitative (Date Added)	

3.3.2. Chart



3.3.3. Evaluation

- **Expressiveness**
 - The amount of content is clearly displayed on each point. Lines of content are represented by the height of the area on the Y axis.
 - The genre trends are fully displayed, represented by different colored lines. The legend is fully displayed for easy recognition by viewers.
 - Support trend analysis, Lines are very effective in showing trends over time. Viewers can see how the amount of listed_in content changes over quarters.
- **Efficiency**
 - Accuracy: Area Chart uses the vertical Position channel (Y axis) to represent the number of contents (Count)
 - Discriminability: Color is a good channel to differentiate categorical categories.
 - Separability: Not considered because no attribute uses 2 channels to represent the same data..
- **Chart analysis**
 - The number of contents increased sharply from 2015, peaked around 2019-2020 (about 160-170 contents per quarter), then gradually decreased in 2021.
 - Popular genre:
 - Through the quarters, we can see that the Documentaries genre has the most fluctuations. Especially in the second quarter of 2018, there were 34 films uploaded to the platform, the most in the period from 2009-2021.
 - The genres “Children & Family Movies”, “Comedies”, “Dramas”, “International Movies” and “Romantic Movies” are always stable but do not account for too large numbers.
 - Trend:
 - Before 2015, the amount of content was very small, mainly “Children & Family Movies”, “Comedies” and “Stand-Up Comedy”.
 - From 2015-2019, genres such as "International TV Shows", "TV Dramas", and "Comedies" increased sharply, reflecting Netflix's strategy of expanding international content and entertainment.
 - After 2020, the amount of content will decrease.

3.4.Domain task 4

3.4.1. Idiom

Idiom	Horizontal Bars	
Data	Rating: C (Categorical)	
	Listed In: C (Categorical)	
	Count: Q (Quantitative)	
Encode	Mark: rectangle	
	Channel	Q: Label (The digits show the quantity)
		C: Color (Color indicates amount of content)

TASK	Compare the amount of content by age rating and content genre. Identify the most popular age rating and content genre pairs.
SCALE	Keys: 12 (Rating), 30 (Listed In) Level: Categorical (for Rating and Listed In), Quantitative (for Count)

3.4.2. Chart

Age-rating category correlations																			
Rating	Listed In																		
	Actio..	Ani..	Ani..	Briti..	Chil..	Class..	Class..	Com..	Crim..	Cult ..	Docu..	Docu..	Dra..	Faith..	Horr..	Inde..	Inter..	Inter..	Kids'..
G	1				50		11	23			9		7				5		
Không xác định			1										1					1	1
NC-17								2					3			4	4		
NR	11			2			4	15	1	3	35	1	41		7	24	60	2	
PG	41	6			305		21	228		5	25		102	18	4	15	27		
PG-13	251	5			18		15	215		16	54		315	12	63	55	140		
R	366						46	233		55	31		595	3	149	303	215	1	
TV-14	230	23	75	49	21	12	25	486	126	7	307	100	780	30	48	154	1,195	498	
TV-G		1		18	57	1	1	30	1		58	25	27	1		3	60	26	48
TV-MA	241	16	66	123	1	11	16	485	403	5	397	190	1,087	3	182	432	1,457	788	
TV-PG	38	22	20	71	106	6	12	171	4	2	200	120	233	22	5	47	341	146	49
TV-Y			2	34	130	1		27					4				1		234
TV-Y7		5	28	9	150	1		60	6				1			3	1	3	263
TV-Y7-F..	2				7			5											1

3.4.3. Evaluation

- **Expressiveness:**
 - Horizontal Bars use color and numbers to represent the quantity of content, with high quantity content also having high correlation which is good for identifying overall trends, but still not good at representing exact values.
 - The number of content based on rating and listed_in pairs is fully displayed on the chart.
 - Horizontal Bars help visually compare the amount of content between pairs of

information and easily recognize the relationship of attributes.

- **Efficiency:**

- Accuracy: Use channel and Color to represent the amount of content, with high amounts indicating a high relationship between the two attributes.
- Discriminability: Age ratings (rating) and genres (listed_in) are clearly distinguished by vertical and horizontal Position on the Y and X axes. Each cell represents a unique pair, with no overlap.
- Separability: Not considered because no attribute uses 2 channels to represent the same data. Rating uses vertical Position, Listed In uses horizontal Position, and Count uses Color. There is no channel overlap causing interference.

- **Chart analysis:**

- Most common pair: The boldest cells with values greater than 1000 are:
 - TV-14 with International Movies
 - TV-MA with International Movies
- See TV-14 and TV-MA rated movies with most content types.
- In the age ratings PG, PG-13 and R, only the genres Action & Adventure, Children & Family Movie, Comedies and Dramas are available, while other genres are almost absent.
- For TV-Y age ratings, TV-Y7 only has Children & Family Movie, Comedies, Kids' TV and TV Comedies.
- For children's age groups, the number and genre of films are limited, so the quantity is small.
- The remaining 2 groups have a rich genre that spans almost all genres.

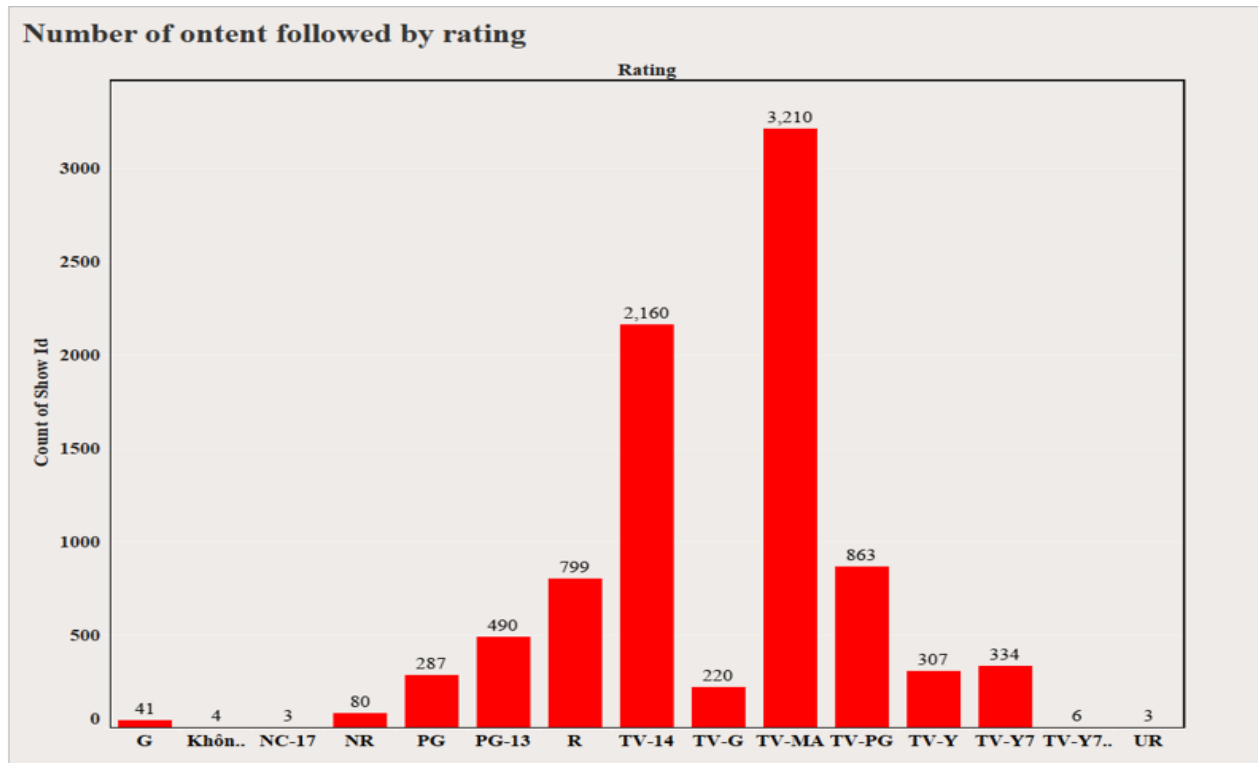
3.5.Domain task 5

3.5.1. Idiom

Idiom	Bar charts	
Data	Rating: C (Categorical)	
	Listed In: C (Categorical)	
Encode	Mark: bar chart	
	Channel	Q: vertical position
		C: horizontal position
TASK	Comparing the amount of content based on rating and audience age	

	Distribution trends by age rating
SCALE	Keys: 14 Level: Categorical

3.5.2. Chart



3.5.3. Evaluation

- **Expressiveness**
 - The amount of content is clearly displayed on top of each column, making it easy for viewers to grasp the exact value.
 - Ratings are displayed in full, without being cropped, making it easy to identify each category.
 - Support trend analysis: viewers can see the amount of content increasing from children (G:41) to teenagers (TV-14: 2160) and adults (TV-MA: 3207)
 - The title should match the content of the chart, helping viewers immediately understand the purpose of the visualization.
- **Efficiency**
 - Accuracy: use position channel (Y axis) to read data accurately.
 - Discriminability: age norms are clearly distinguished by their position on the X-axis.
 - Separability: not considered because no attribute uses 2 channels

- **Chart analysis**
 - TV-MA is the most popular rating type with 3,207 content, followed by TV-14 (2,160) and TV-PG (863).
 - Other categories like UR and NC-17 are very few (3 each) showing that this content is not popular on Netflix.
 - Grouping:
 - Children's group (G, TV-G, TV-Y, TV-Y7, TV-Y7-FV): 908 contents.
 - Teen (PG, TV-PG, PG-13, TV-14): 3,800 titles.
 - Adult (R, TV-MA, NC-17): 4,009 entries.
 - Unranked group (NR, UR): 83 contents.

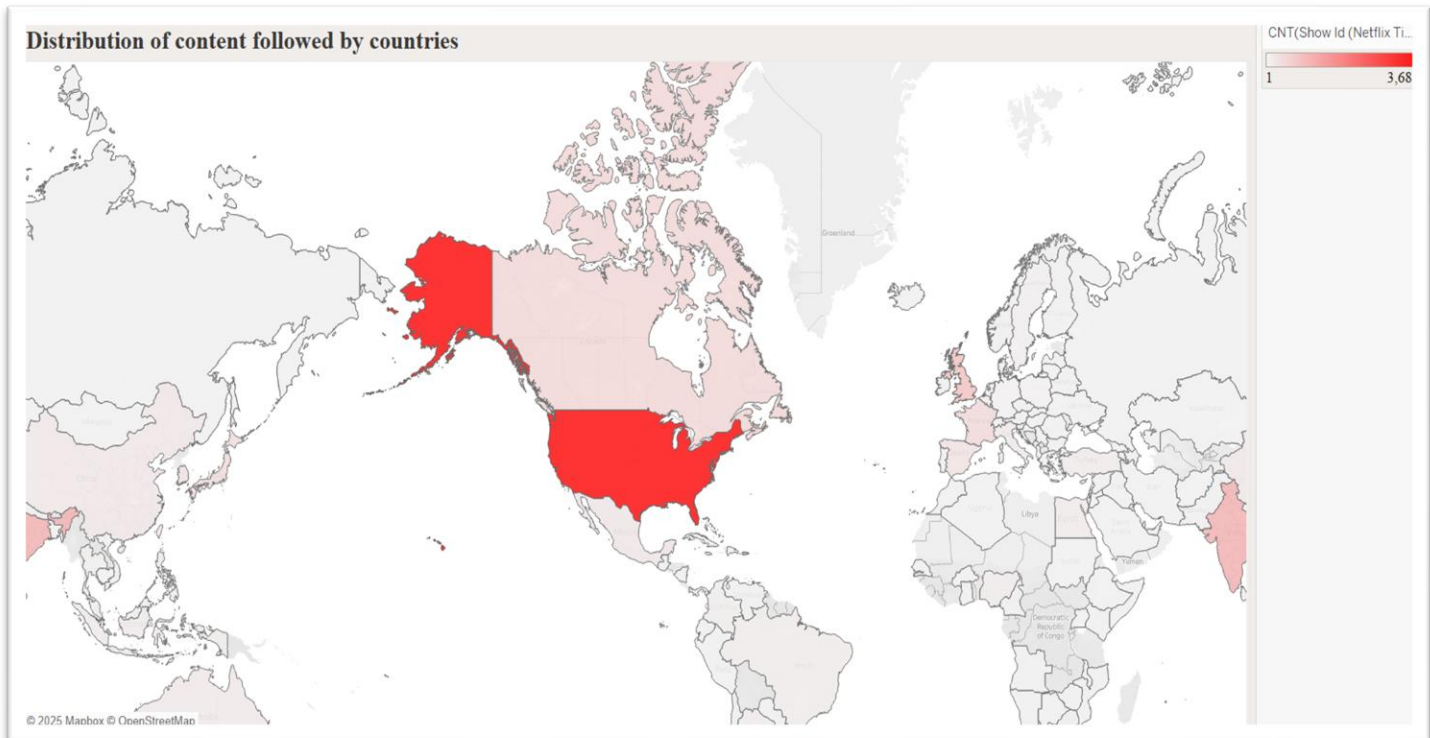
3.6.Domain task 6

3.6.1. Idiom

Idiom	Line charts	
Data	Country: C	
	Latitude (generated): Q1	
	Longitude (generated): Q2	
	Count: Q	
Encode	Mark: line	
	Channel	Q1: vertical position
		C: color
		Q: title
		Q2: horizontal position
TASK	Compare content volume across countries across geographic regions	
SCALE	Level: Nominal, Interval	

3.6.2. Chart

3.6.3. Evaluation



- **Expressiveness**
 - The map chart clearly shows the distribution of Netflix content quantity by country, with color coding the number of titles (CNT).
 - The title “Distribution of Netflix Content by Country of Production (2008-2021)” accurately reflects the content of the chart, helping viewers understand the purpose of the visualization.
 - Country labels are displayed directly on the map, encoded by geographic location and color, helping viewers easily identify countries and the corresponding amount of content.
- **Effectiveness**
 - **Accuracy:** Use location channels (Latitude, Longitude) on the map to locate countries precisely, and colors to represent content quantity (CNT), making it easy for viewers to compare.
 - **Discriminability:** Use a color scale from light yellow to dark red to differentiate the amount of content, easily identifying countries with a lot of content (dark red) and little content (light yellow).
 - **Separability:** Not considered because there is no attribute that uses 2 channels to represent at the same time.
- **Chart analysis**
 - Shows the geographic distribution of Netflix content, with the United States (2,818 titles) and India (912 titles) leading in terms of content volume.

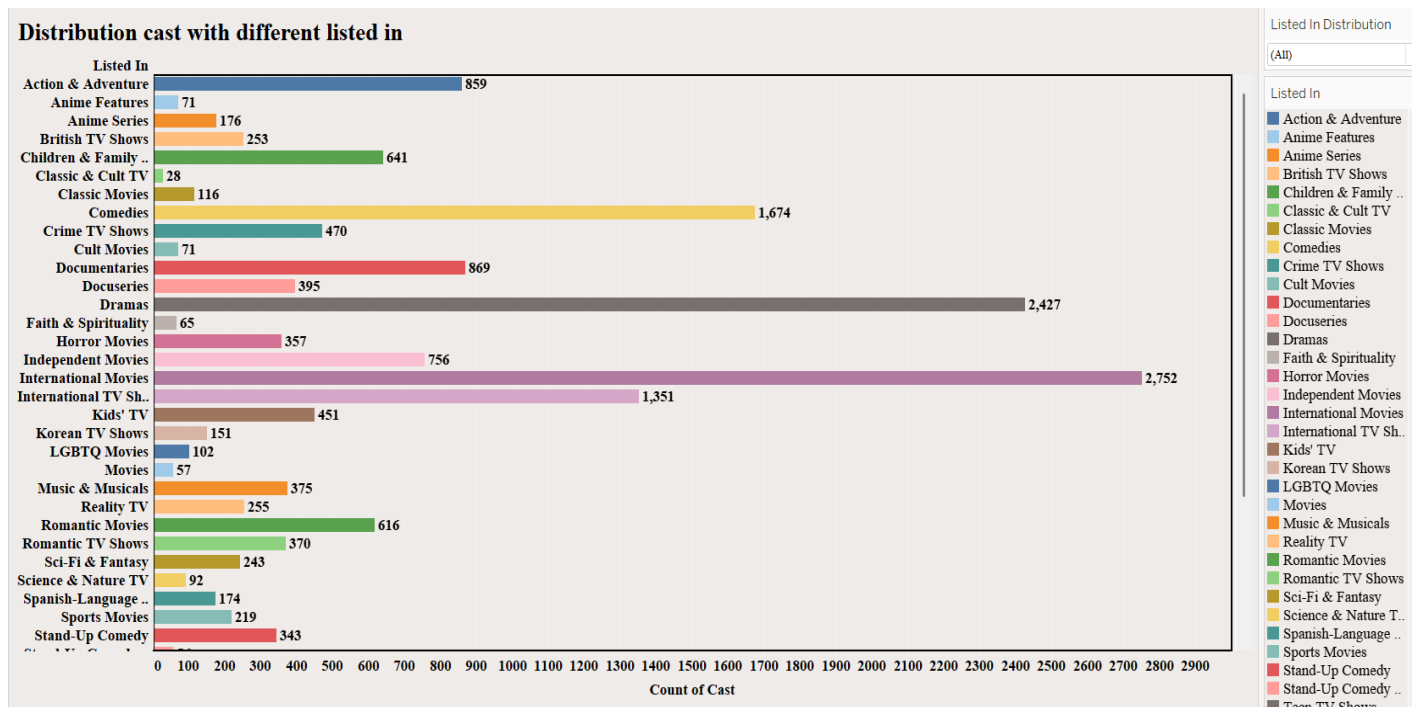
- North America (United States, Canada, Mexico) is the region with the largest content output, while South Asia (India, Pakistan, Bangladesh) and East Asia (Korea, Japan, Hong Kong) also contribute significantly. Smaller countries such as Guatemala and Cambodia have very little content (1 title).

3.7.Domain task 7

3.7.1. Idiom

Idiom	Bar charts	
Data	Cast: Q	
	Listed in: C	
Encode	Mark: bar	
	Channel	Q: horizontal position
		C: color
		C: vertical position
TASK	Distribution of the number of actors by film genre	
SCALE	Keys: 42 Level: Nominal	

3.7.2. Chart



3.7.3. Evaluation

- Expressiveness
 - The bar chart clearly shows the number of actors playing each genre.

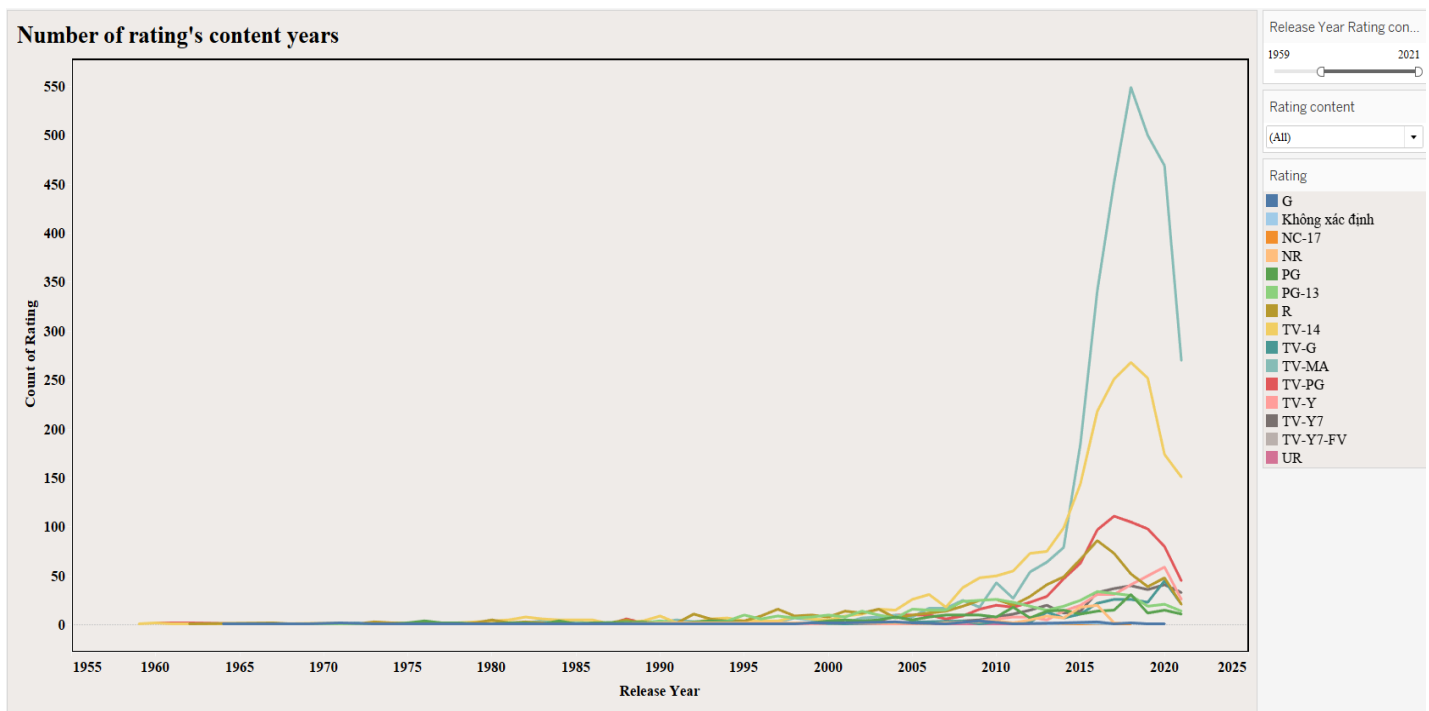
- The title accurately reflects the content of the chart, letting the viewer know the purpose of the visual image.
- The labels “Listed in” and “count (cast)” clearly express what the user needs to do.
- **Efficiency**
 - Accuracy: use position channel (Y axis) to distinguish genre accurately.
 - Discriminability: Use color to distinguish film genres.
 - Separability: not considered because there is no attribute that uses 2 channels to represent.
- **Chart analysis**
 - Clearly show the number of actors participating in each film genre.
 - International movies have the highest total number of actors while TV shows have the lowest number.

3.8.Domain task 8

3.8.1. Idiom

Idiom	Line charts	
Data	Rating: C	
	Release Year: Q	
Encode	Mark: line	
	Channel	C: vertical position
		C: color
		Q: horizontal position
TASK	Distribution and trends of age-rated movies released by year.	
SCALE	Keys: 18	
	Level: Nominal	

3.8.2. Chart



3.8.3. Evaluation

- **Expressiveness**
 - The bar chart clearly shows the number of age-rated movies released by year.
 - The title accurately reflects the content of the chart, letting the viewer know the purpose of the visual image.
 - The “rating” label helps users easily distinguish which age classification.
- **Efficiency**
 - Accuracy: use position channel (Y axis) to distinguish age classification.
 - Discriminability: Use color to differentiate age classification.
 - Separability: Not considered because there is no attribute that uses 2 channels for representation.
- **Chart analysis**
 - Clearly shows the trend of age-rated movies released each year.
 - TV-MA tends to increase and decrease slightly but leads each year of release. Meanwhile, the ages with the titles “66 min”, “74 min”, “84 min” are only released in one year and are the number of films with the lowest age rating.

